

Economic Dispatch with Wind Power Integration in an IEEE 24-Bus Test System

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Abstract—The growth of generation technologies has enabled the incorporation of renewable resources into electric power systems. This integration requires studies of power flow, optimal dispatch, and network operating limits. This paper reformulates an economic dispatch problem as a DC optimal power flow (DC-OPF) model for an IEEE 24-bus test system. Two operating cases are evaluated: a conventional generation case and a second case that includes a wind generator connected at bus 3. The optimization model was implemented in Julia using JuMP and the GLPK linear programming solver. The results show that wind generation preserves the power balance and line-flow constraints while reducing total generation cost. The comparison indicates a reduction of approximately \$1,952 in average hourly cost and \$1,600 in maximum hourly cost when the wind generator is included.

Index Terms—renewable generation, power systems, DC optimal power flow, economic dispatch, wind power, Julia, JuMP.

I. INTRODUCTION

Electric power systems are complex because of their size, operating constraints, and the stochastic behavior of demand and generation. As demand grows, system operators must continuously adapt generation schedules to preserve reliability, security, and real-time power balance [1], [2]. In Ecuador, this operational role is performed by CENACE, which determines hourly dispatch schedules from variable generation costs while respecting technical constraints.

Economic dispatch and optimal power flow (OPF) address the need to supply load at the lowest possible cost. Economic dispatch determines the least-cost combination of generating units, whereas OPF incorporates the network model and operating limits. These problems become especially relevant when renewable sources are added, because renewable generation generally has a lower marginal cost than conventional generation but introduces variability and uncertainty.

This work develops an economic dispatch algorithm for an IEEE 24-bus test system and evaluates how the dispatch profile changes when a wind generator is added. The purpose is to verify whether the renewable source can reduce operating cost while maintaining demand coverage and respecting system constraints.

II. PROBLEM STATEMENT

The study considers an IEEE 24-bus test system (Fig. 1). A wind generator is added at bus 3 and compared against the

base case with only conventional generators.

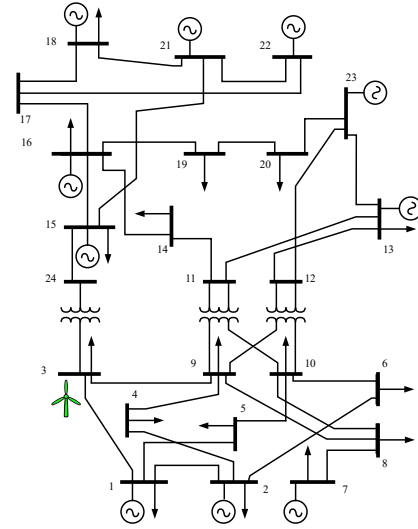


Fig. 1. IEEE 24-bus test system

The conventional units, their bus locations, and active-power limits are listed in Table I. The wind unit is treated as a scheduled renewable source. Since wind output depends on wind speed and turbine characteristics, no fixed minimum and maximum conventional limits are assigned to it. Instead, the model uses the available hourly wind dispatch profile supplied with the original data set (Fig 2).

TABLE I
CONVENTIONAL GENERATOR DATA

Unit	Bus	$P_i^{\max}$ (MW)	$P_i^{\min}$ (MW)
1	1	152	30.40
2	2	152	30.40
3	7	350	75.00
4	13	591	206.85
5	15	60	12.00
6	15	155	54.25
7	16	155	54.25
8	18	400	100.00
9	21	400	100.00
10	22	300	300.00
11	23	310	108.50
12	23	350	140.00

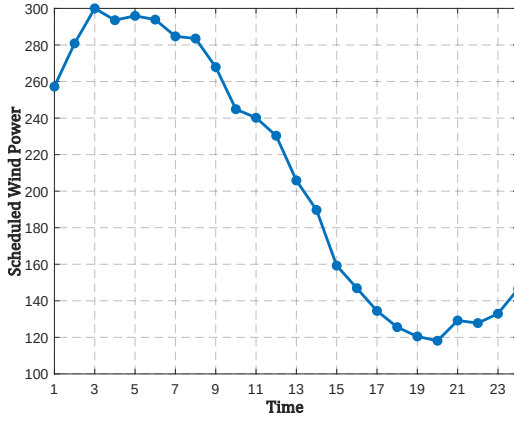


Fig. 2. Scheduled wind power dispatch

For economic dispatch, the objective is to minimize the total generation cost. The unit generation costs used by the model are shown in Table II. The wind generator and generator 10 are assigned zero cost in the study data.

TABLE II
GENERATION COSTS

Unit	C_i (\$/MWh)
1	13.32
2	13.32
3	20.70
4	20.93
5	26.11
6	10.52
7	10.52
8	6.02
9	5.47
10	0.00
11	10.52
12	10.89
Wind	0.00

The hourly system demand is provided in Table III, and the load distribution among buses is provided in Table IV. These data define the active-power demand to be supplied at each hour and at each load bus.

TABLE III
HOURLY DEMAND PROFILE

Hour	Demand (MW)	Hour	Demand (MW)
1	1775.835	13	2517.975
2	1669.815	14	2517.975
3	1590.300	15	2464.965
4	1563.795	16	2464.965
5	1563.795	17	2623.995
6	1590.300	18	2650.500
7	1961.370	19	2650.500
8	2279.430	20	2544.480
9	2517.975	21	2411.955
10	2544.480	22	2199.915
11	2544.480	23	1934.865
12	2517.975	24	1669.815

TABLE IV
LOAD LOCATION AND DEMAND DISTRIBUTION

Load	Bus	% Demand	Load	Bus	% Demand
1	1	3.8	10	10	6.8
2	2	3.4	11	13	9.3
3	3	6.3	12	14	6.8
4	4	2.6	13	15	11.1
5	5	2.5	14	16	3.5
6	6	4.8	15	18	11.7
7	7	4.4	16	19	6.4
8	8	6.0	17	20	4.5
9	9	6.1	—	—	—

Transmission-line reactances and capacity limits are shown in Table V. They are used to constrain the active-power flow between connected buses.

III. DC OPTIMAL POWER FLOW FORMULATION

The DC-OPF formulation is used because it provides a linear approximation of the AC power-flow problem. The model neglects active-power losses, assumes voltage magnitudes close to 1 p.u., and assumes small voltage-angle differences between adjacent buses [3]. Under these assumptions, the optimization focuses on active-power dispatch, bus voltage angles, and line-flow limits.

The objective function minimizes total generation cost over the generating units:

$$\min \sum_{i=1}^{N_G} C_i P_{g,i}, \quad (1)$$

where C_i is the marginal generation cost of unit i , $P_{g,i}$ is its active-power output, and N_G is the number of generating units.

A. Power Balance Constraint

At each bus, the generated power plus the net power exchanged through adjacent lines must satisfy the local load:

$$P_{g,i} + \sum_{j:(i,j) \in \mathcal{L}} P_{ij} = P_{d,i}, \quad \forall i \in \mathcal{N}, \quad (2)$$

where $P_{d,i}$ is the demand at bus i , $\mathcal{N}$ is the bus set, and $\mathcal{L}$ is the line set. In the DC model, the line flow is expressed as

$$P_{ij} = \frac{\theta_i - \theta_j}{X_{ij}}, \quad (3)$$

where θ_i and θ_j are voltage angles and X_{ij} is the line reactance.

B. Slack Bus Constraint

A reference or slack bus is required to define all bus voltage angles. The slack-bus angle is fixed as

$$\theta_{\text{slack}} = 0. \quad (4)$$

TABLE V
TRANSMISSION LINE PARAMETERS

From	To	X (p.u.)	Cap. (MVA)	From	To	X (p.u.)	Cap. (MVA)
1	2	0.0146	175	11	13	0.0488	500
1	3	0.2253	175	11	14	0.0426	500
1	5	0.0907	350	12	13	0.0488	500
2	4	0.1356	175	12	23	0.0985	500
2	6	0.2050	175	13	23	0.0884	500
3	9	0.1271	175	14	16	0.0594	500
3	24	0.0840	400	15	16	0.0172	500
4	9	0.1110	175	15	21	0.0249	1000
5	10	0.0940	350	15	24	0.0529	500
6	10	0.0642	175	16	17	0.0263	500
7	8	0.0652	350	16	19	0.0234	500
8	9	0.1762	175	17	18	0.0143	500
8	10	0.1762	175	17	22	0.1069	500
9	11	0.0840	400	18	21	0.0132	1000
9	12	0.0840	400	19	20	0.0203	1000
10	11	0.0840	400	20	23	0.0112	1000
10	12	0.0840	400	21	22	0.0692	500

C. Generation Limits

Each conventional generator is constrained by its minimum and maximum active-power output:

$$P_{g,i}^{\min} \leq P_{g,i} \leq P_{g,i}^{\max}, \quad \forall i \in \mathcal{G}. \quad (5)$$

For the wind unit, the generated power is fixed to the scheduled hourly wind profile.

D. Line Capacity Limits

Line-flow constraints limit the active power transferred through each transmission element:

$$-P_{ij}^{\max} \leq \frac{\theta_i - \theta_j}{X_{ij}} \leq P_{ij}^{\max}, \quad \forall (i, j) \in \mathcal{L}. \quad (6)$$

These limits prevent overloads and ensure that the dispatch remains technically feasible.

IV. IMPLEMENTATION

The model was implemented in a Julia Jupyter Notebook. The main packages used in the project are summarized in Table VI.

TABLE VI
LIBRARIES USED IN THE PROJECT

Library	Purpose
JuMP	Optimization modeling package for Julia.
GLPK	Linear programming solver used by the DC-OPF model.
MAT	Data export package for saving results in MATLAB format.
Plots	Plotting package for result visualization.

Because the goal is to analyze hourly economic dispatch over one day, the optimization is solved for 24 time periods. For each hour, the algorithm creates a JuMP model, defines the decision variables, applies the objective function in (1), and enforces the constraints in (2)–(6). The dispatch variable depends on generator, bus, and time, whereas the angle variable depends on bus and time.

The power-balance constraint was implemented by considering four bus cases: buses with generation, lines, and load; buses with generation and lines but no load; buses with lines and load but no generation; and buses with lines but neither load nor generation. Special treatment is required at bus 15, where two conventional generators are connected. After solving each hourly optimization problem, the results are saved in arrays summarized in Table VII.

TABLE VII
STORED RESULT ARRAYS

Array	Description
Dispatch	$G \times T$ array containing dispatched power for generator g at time t .
Angle	$B \times T$ array containing voltage angles for bus b at time t .
Balance	$B \times T$ array containing bus power-balance results.
Line	$L \times T$ array containing active-power flow through line l at time t .
Cost	$1 \times T$ array containing the minimized objective-function value at each time t .

The resulting arrays were exported to MATLAB format to support additional plotting and inspection [5].

V. RESULTS

A. OPF Without Wind Generation

The first scenario considers only the 12 conventional generators. The generator outputs satisfy the generation limits in Table I. The main results for this scenario are shown through the generator contribution heatmap in Fig. 3, the line-loading heatmap in Fig. 4, the demand-compliance plot in Fig. 5, and the hourly cost function in Fig. 6.

As shown in Fig. 3, the dispatch results are presented as a heatmap of percentages, where each value represents the contribution of each conventional unit to the total hourly demand. Fig. 4 reports the loading percentage of each transmission line with respect to its capacity, making it possible to verify that the network operates within its limits. Fig. 5 compares

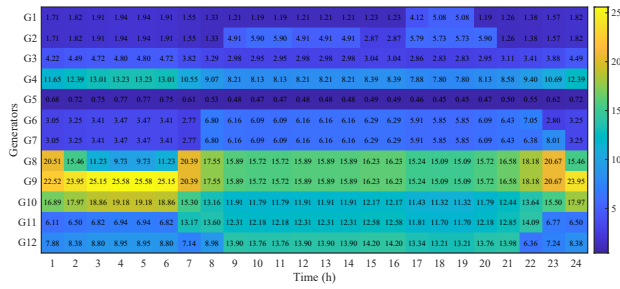


Fig. 3. Generator contribution to hourly demand - No Wind generation

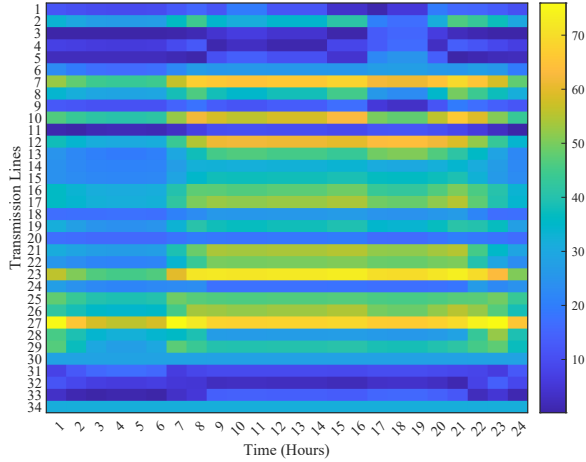


Fig. 4. Transmission Lines Loadability Percentage Profile - No Wind Generation

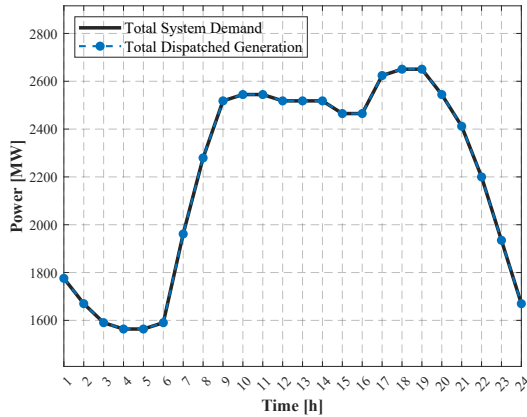


Fig. 5. System Demand Compliance Profile - No Wind Generation

total generation with the hourly demand profile and confirms that the conventional units satisfy the required load at every time period. Finally, Fig. 6 shows the hourly operating cost obtained from the minimized objective function for the no-wind scenario.

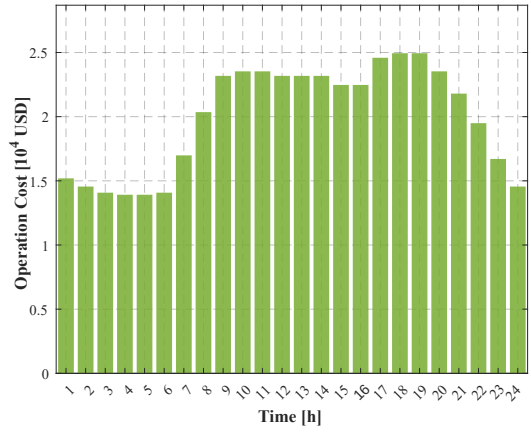


Fig. 6. Hourly System Operation Cost Profile - No Wind Generation

B. OPF With Wind Generation

The second scenario includes the wind generator at bus 3. The first verification is that the wind unit follows the scheduled wind profile. The remaining conventional generators then adapt their outputs to preserve power balance and minimize cost. The main results for this scenario are shown through the generator contribution heatmap in Fig. 7, the line-loading heatmap in Fig. 8, the demand-compliance plot in Fig. 9, and the hourly cost function in Fig. 10.

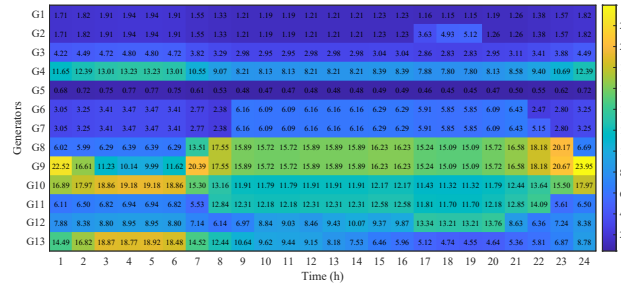


Fig. 7. Generator contribution to hourly demand - Wind generation

As shown in Fig. 7, the dispatch results are presented as percentage contributions to the hourly demand, including the participation of the wind generator. This heatmap shows how the scheduled wind source changes the share supplied by the conventional units over the day. Fig. 8 reports the transmission-line loading percentages after the wind unit is connected, confirming that the network remains within its line-capacity limits. Fig. 9 shows that total generation continues to match the hourly demand profile, so the model remains feasible after adding wind generation. Finally, Fig. 10 presents the hourly operating cost for the wind scenario; because the wind generator has zero assigned cost, the objective-function values decrease relative to the conventional-only case.

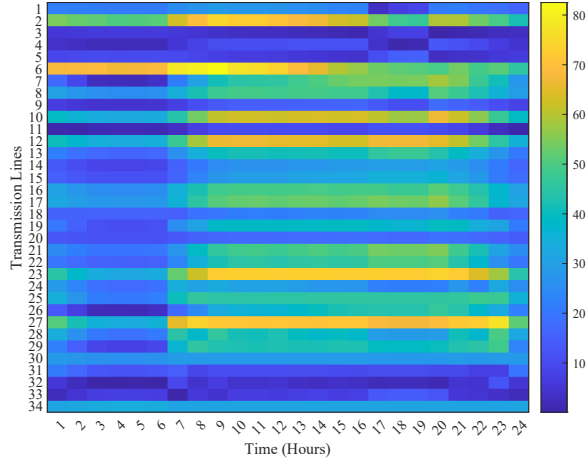


Fig. 8. Transmission Lines Loadability Percentage Profile - Wind Generation

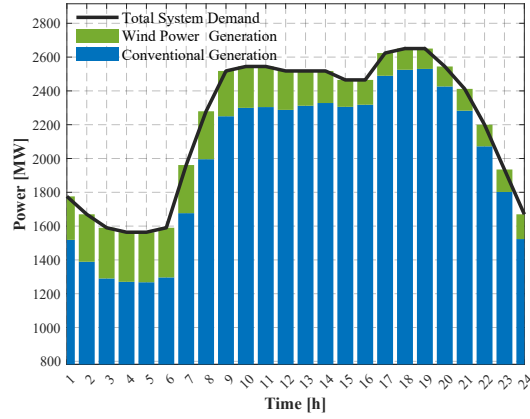


Fig. 9. System Demand Compliance Profile - Wind Generation

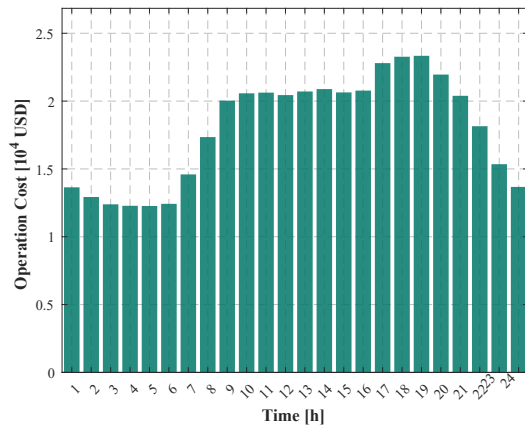


Fig. 10. Hourly System Operation Cost Profile - Wind Generation

C. Scenario Comparison

The scenario comparison is based on three displacement plots: the generator dispatch difference in Fig. 11, the transmission-line loadability difference in Fig. 12, and the objective-function comparison in Fig. 13. These figures provide a more compact view of how wind integration modifies generation, network loading, and operating cost.

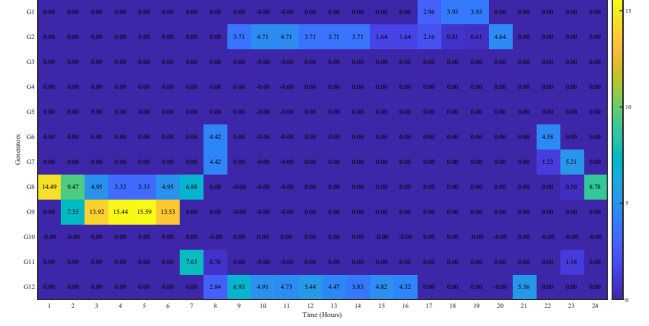


Fig. 11. Generator Dispatch Displacement Level (With Wind vs. No Wind)

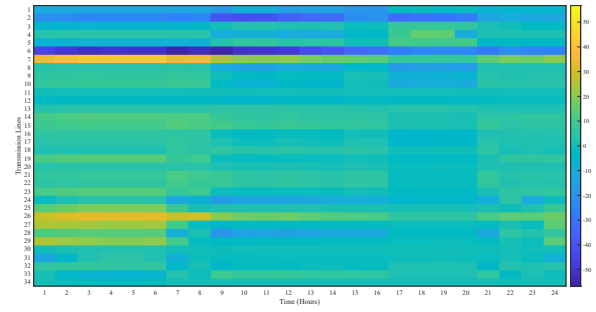


Fig. 12. Transmission Lines Loadability Displacement Profile (With Wind vs. No Wind)

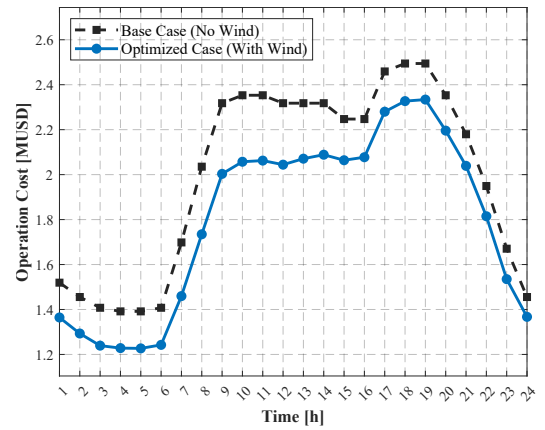


Fig. 13. Economic Dispatch Objective Function Comparison

Fig. 11 shows the difference between the no-wind and wind generator-contribution heatmaps. Positive values represent a reduction in the contribution of a unit after adding wind generation. The largest displacement appears in generators 8

TABLE VIII
COST COMPARISON BETWEEN SCENARIOS

Analysis parameter	Without wind	With wind
Maximum cost	\$24,940	\$23,340
Minimum cost	\$13,920	\$12,280
Average cost	\$19,931	\$17,979

and 9 during the low-demand hours, while generators 2 and 12 also reduce their contribution during several medium- and high-demand periods. Generators 3, 4, 5, and 10 remain nearly unchanged, which is consistent with the optimization results: the expensive units stay close to their minimum limits, and generator 10 remains fully dispatched because it has zero assigned cost.

Fig. 12 compares the transmission-line loadability profiles between scenarios. Negative blue values indicate that the line loadability increases when wind generation is included, while positive yellow values indicate a reduction in line loading. The figure shows that wind integration redistributes power flows across the network rather than uniformly reducing all line loadings. Some corridors become less loaded, whereas others carry more power because the additional generation is injected at bus 3 and changes the DC power-flow pattern.

Fig. 13 compares the hourly operating cost in both scenarios. The wind case remains below the no-wind case throughout the day, confirming the economic benefit of using the zero-cost wind source. The maximum cost occurs at hours 18 and 19 in both scenarios, coinciding with the maximum demand of 2650.5 MW. The minimum cost occurs at hours 4 and 5, when demand reaches 1563.795 MW. These coincidences confirm that the cost profiles remain consistent with the hourly demand profile. Table VIII summarizes the resulting maximum, minimum, and average cost values.

VI. CONCLUSIONS

This paper presented a DC-OPF-based economic dispatch model for an IEEE 24-bus test system with and without wind generation. The model was implemented in Julia using JuMP and GLPK, and it considered generator limits, line-flow limits, slack-bus reference, and hourly bus power balance.

The results show that adding a wind generator at bus 3 reduces the objective-function cost while maintaining feasible operation. The average hourly cost decreases from \$19,931 to \$17,979, yielding an average reduction of \$1,952. The maximum hourly cost decreases from \$24,940 to \$23,340, yielding a reduction of \$1,600. The demand is satisfied in every hour in both scenarios, and the line-flow constraints remain within their allowed limits.

The dispatch comparison confirms that the model prioritizes low-cost resources. Generator 10 and the wind generator are used preferentially because they have zero assigned cost, while expensive units remain near their minimum operating limits except when additional power is required. This behavior validates the economic logic of the optimization algorithm.

The main limitation of the study is that wind generation is represented by a predetermined dispatch profile. In real systems, wind power is uncertain and depends on wind-speed measurements, turbine power curves, and forecasting accuracy. Future work should incorporate wind uncertainty, historical wind data, and unit commitment constraints to represent a more realistic operational planning problem.

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